

Assignment 9

Name: _____

Student ID: _____

9 problems. Show all of your work.

Problem 1.

Find an equation for the tangent plane to the graph of $f(x, y) = \cos(xy)$ passing through the point $\left(\frac{\pi}{2}, 1, 0\right)$.

Problem 2.

Find a parametric equation for a line contained in the tangent plane you found in the previous problem. (Any line will suffice.)

Problem 3.

Find the equation of the tangent plane at the point $P_0(2, 0, 2)$ on the surface $2z - x^2 = 0$.

Problem 4. True or False

Any tangent plane to a graph must meet that graph at exactly one point.

Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Problem 5.

Let S be a sphere centered at the origin in $\mathbb{R}^3$, and consider any point P on S . Show that the vector $\vec{OP}$ is orthogonal to the tangent plane to S at P .

Problem 6.

Find the equation of the tangent plane to the sphere $x^2 + y^2 + z^2 = 14$ at $(1, 2, 3)$.

Problem 7.

The tangent plane to $z = f(x, y)$ at the point $(1, 2, 9)$ is $z = 3x + 5y - 4$. Find $\nabla f(1, 2)$.

Problem 8.

Let $f(x, y) = 2 + \ln(1 + 3x - 4y)$. Use a linearization to approximate $f(0.1, 0.2)$.

Problem 9.

Let

$$g(x, y) = \frac{\sqrt{x}}{\sqrt[3]{y}},$$

and note that $g(36, 8) = 3$. Use an appropriate partial derivative to approximate $g(36, 6)$.